Miraj Pudasaini

Data Engineer
London, England [SE2 0PA] | 07919445314 pudasainimiraj@outlook.com

Work Experience

Sr. Data Engineer (Airflow, Python, SQL, ETL, Databricks, Azure). June 2025 - Present
I Tech Care LTD. | London

- Design and build high-performance data pipelines utilizing Databricks and Apache Spark connectors in Airflow to extract, transform, and load *Prices and Market* ticker from well-known **stock exchanges** into Azure Data Lake Storage and related Azure services
- Develop and maintain secure data warehouses and Lakehouse's through robust data models, comprehensive quality checks, and stringent governance practices
- Deigned and deployed data infrastructures to improve productivity of data team by 40% by aiding collaboration on Data Quality Analysis and Data ingestion.
- Designed and deployed numerous vertical AI agent performing ***RAG*** on *EODH* time series ticker to automate financial market data analysis.
- Spearheaded implementation of hybrid B25 and Titan-embedding search over millions of financial series, gated by nDCG@10 and P@12.
- Facilitated automated reporting of the data and data process health to detect failure and anomaly before they occur to save clients resources.

Data Engineer (Dragster, Snowflake, Excel, Python, SQL, AWS) Feb 2023 – May 2025
Zola Analytics | London

- Engineered ETL data-pipeline, replacing Excel workflows with a modern tech/data stack - Dragster, and custom Python/Pandas solutions increasing ingestion frequency by 60% simultaneously improving the Monitoring and Alerting to improve organizational data process by 80%.
- Optimized Zola's powerful data platform built in python, SQL alchemy, neo4j and timescale DB to improve security and transaction time 70% and saved on AWS bills each month in storage cost by applying best database design patterns.
- Saved company resource multiple times by triaging AWS compute, detecting anomaly in cloud infrastructure and serverless architecture by accurately diagnosing memory leaks in system and deploying quick fixes.
- Designed and implemented bespoke API clients to orchestrate migration of huge databases like OECD, IMF, BIS, FRED etc. and dimension modelled the time-series data to fit Zola data platform.
- Collaborated with Machine Learning and AI team to design and deploy an agentic Large Language model to perform RAG on Macro Time Series Data to automate charts creation.

Full stack Software Engineer April 2019 – June 2020
Green codes | Kathmandu Nepal

- Instrumental in modernizing a legacy Java based internal CRM application to Node.js/React/Next, eliminating the bottlenecks and latency in Api responses and making company CRM and dev tools 12% faster.
- Collaborated as a part of a customer centric agile team to develop and maintain RESTful APIs written in Spring Boot deployed over AWS.
- Spearheaded implementation of AWS DevOps pipeline to switch to Serverless optimizing ci/cd operation cost to the company by 20%.
- Improved SQL queries to make user retrieval process consume 20% less database resources.

Projects

ContextTrace (pudasainimiraj.github.io/ContextTrace)— semantic search over GitHub discussions (solo, Oct 2026) · TypeScript, React, Cloudflare Workers, Workers AI, Supabase pgvector

- Built a retrieval service that answers questions about a repository with passages linked to the source issue or PR comment.
- Designed an incremental, idempotent ETL on Cloudflare Queues with deterministic passage ids and content hashes.
- Ran the service entirely on free tiers with budget ledgers, job leases and retries; cut transform CPU 4 to 6x.
- Measured quality on a frozen 25-question set: Hit@3 0.65, thread Hit@5 0.95; the keyword baseline scored 0.05

Education

York St. John's University *(awarded Upper Second Class, 69.6%)* Feb 2021-Feb 2023
London, United Kingdom

Master of Science | Computer Science

University of Wolverhampton *(awarded First Class w/ Honors,75.8%)* Sept 2017-July 2020
(Herald College Kathmandu). Kathmandu, Nepal

Bachelor of Science (Hons.) | Computer Science